Math Behind the 85% Ashflow Alpha Portfolio Allocation

1. Set-up

The trading budget is

$$B = 1{,}000{,}000.$$

Let

$$x_i = \text{percentage of budget allocated to product } i,$$

where x_i is written in percentage points. For example, $x_i = 10$ means 10% of the budget.

The dollar investment in product i is therefore

$$I_i = B\frac{x_i}{100}.$$

The fee used for the calculation was interpreted as a quadratic product-level fee:

$$F_i = B\left(\frac{x_i}{100}\right)^2.$$

So the total fee is

$$F = B\sum_i \left(\frac{x_i}{100}\right)^2.$$

This matters because the fee grows with the square of position size. Doubling one position doubles exposure, but quadruples the fee:

$$F(2x) = B\left(\frac{2x}{100}\right)^2 = 4B\left(\frac{x}{100}\right)^2.$$

2. Objective function

Let a_i be the expected return edge for product i, after choosing the correct direction. For a buy, a_i is the expected positive return from going long. For a sell, a_i is the expected positive return from going short.

Using

$$y_i = \frac{x_i}{100},$$

the expected net PnL contribution from product i is approximately

$$\Pi_i = B\left(a_i y_i - y_i^2\right).$$

Therefore the portfolio objective is

$$\max_{y_i} \Pi = B \sum_i \left(a_i y_i - y_i^2\right),$$

subject to

$$\sum_i y_i \le 1, \qquad y_i \ge 0.$$

Unused budget has no direct value, but using extra budget is only good if the added expected return is larger than the extra fee.

3. Marginal benefit versus marginal fee

For one product,

$$\Pi_i = B(a_i y_i - y_i^2).$$

The marginal expected net PnL is

$$\frac{d\Pi_i}{dy_i} = B(a_i - 2y_i).$$

Adding more to product i is worthwhile only when

$$a_i > 2y_i.$$

At the unconstrained optimum,

$$a_i - 2y_i = 0,$$

so

$$y_i^* = \frac{a_i}{2}.$$

This is the key rule: a position should only be increased until its expected edge is roughly equal to its marginal fee penalty.

4. Why not force the full 100%?

Suppose an extra allocation Δy is added to a product already sized at y_i. The extra net value is approximately

$$\Delta\Pi_i \approx B(a_i - 2y_i)\Delta y.$$

If the remaining signals are weak, uncertain, or already crowded, then a_i may be smaller than $2y_i$. In that case,

$$\Delta\Pi_i < 0,$$

meaning the extra allocation lowers expected PnL after fees.

That is why using 85% can be rational even though unused budget expires. The unused 15% is not being saved for value; it is being avoided because the expected marginal after-fee return is not strong enough.

5. Chosen allocation and fee calculation

The proposed portfolio was:

Product	Direction	Allocation x_i	Fee contribution
Obsidian cutlery	Buy	8%	6,400
Pyroflex cells	Sell	12%	14,400
Thermalite core	Buy	13%	16,900
Lava cake	Sell	13%	16,900
Magma ink	Buy	9%	8,100
Scoria paste	Buy	5%	2,500
Ashes of the Phoenix	Sell	7%	4,900
Volcanic incense	Buy	4%	1,600
Sulfur reactor	Buy	14%	19,600
Total		85%	91,300

The total fee is

$$F = 1{,}000{,}000 \left(0.08^2 + 0.12^2 + 0.13^2 + 0.13^2 + 0.09^2 + 0.05^2 + 0.07^2 + 0.04^2 + 0.14^2\right).$$

$$F = 91{,}300.$$

6. Why the allocation is spread across products

The fee is convex. For a fixed total exposure, spreading across multiple good signals is cheaper than concentrating everything in one product.

For example, putting 20% into one product costs

$$1{,}000{,}000(0.20)^2 = 40{,}000.$$

Splitting the same 20% equally across two products costs

$$1{,}000{,}000(0.10^2 + 0.10^2) = 20{,}000.$$

So diversification has a direct fee advantage, but only across products where the news signal still gives positive expected edge.

7. Interpretation of the selected sizes

The strongest allocations went to signals with clearer one-day catalysts:

- Sulfur reactor: index inclusion creates likely mechanical buying pressure.
- Thermalite core: demand forecast sharply increases.
- Lava cake: health review and shelf removal are clearly negative.

- Pyroflex cells: the end of the tax cut is negative for demand.

Smaller allocations were used where the signal was weaker, more speculative, or more likely to be crowded:

- Scoria paste: bullish stockpiling narrative, but based on celebrity-style hype.
- Volcanic incense: social attention surge, but potentially already overextended.
- Ashes of the Phoenix: negative public backlash, but less directly tied to the tradable good's immediate price.

8. Final conclusion

The 85% allocation was intentional. The decision comes from comparing expected marginal return against the quadratic fee penalty:

$$\text{Add more only if} \qquad a_i > 2y_i.$$

Because the last 15% of budget would likely need to go into weaker or more crowded signals, the expected after-fee value of using it was judged lower than the quadratic fee cost. Therefore, the portfolio uses the strongest signals while avoiding low-confidence marginal exposure.